

Power System Project

Initials & Surname:

Date:

Project

TD Masinga

27 October 2024

**SYSTEM PLANNING FOR GENERATION
RETIREMENT**

Contents

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1. Introduction and background:

Overview:

The modern power grid faces unprecedented challenges in ensuring efficient, reliable, and sustainable electricity delivery.

As global energy demand continues to rise, power systems must adapt to integrate diverse generation sources, manage increasing load complexity, and maintain stability in the face of dynamic operational conditions. One critical aspect of achieving these goals is the strategic application of power system compensation techniques.

Compensation methods, including shunt and series compensation, play a vital role in mitigating voltage instability, reducing transmission losses, and enhancing overall system performance.

Moreover, the integration of renewable energy sources offers a promising avenue for sustainable voltage support and reduced carbon emissions. However, optimizing compensation strategies requires a deep understanding of their interactions and impacts on power system dynamics.

The purpose of this project is to study a five-bus power system network with two generators and five loads connected by transmission lines.

The project aims to evaluate system performance under different compensation scenarios using PowerWorld Simulator to measure the impact of each compensation method on the system's efficiency and stability.

Objectives:

- To understand the baseline performance of a power system network without any compensation.
- To investigate the effects of shunt and series compensation on voltage stability and power transfer efficiency.
- To explore renewable energy integration as an alternative to traditional compensation methods.

Research Significance:

This project addresses key challenges in modern power systems, such as voltage stability, efficient power transfer, and sustainable energy sources. The findings will provide valuable insights for system engineers working on grid stability, optimization, and transitioning to renewable energy sources..

The findings will guide decisions on compensation techniques, emphasizing the importance of understanding each method's context before application.

2. Case Study System Modelling and Implementation

Effective power system compensation requires a thorough understanding of the underlying system dynamics.

To investigate the impact of different compensation methods, a comprehensive system model was developed using the PowerWorld Simulator.

The five-bus power system network, comprising two generators and five loads connected by transmission lines, serves as the foundation for this study.

The system model incorporates critical parameters such as generator capacities, load demands, transmission line impedances, and bus configurations.

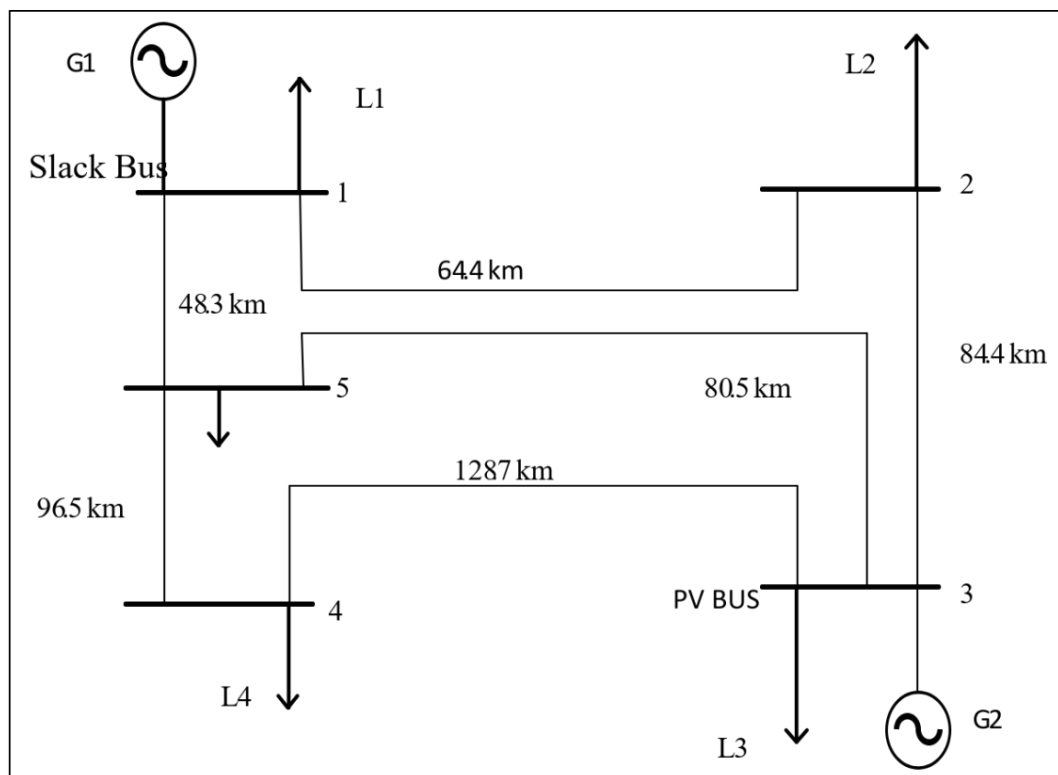


Figure 1: Five Bus Power System Network .

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Generation and Load Parameters for the above system are given as follows

Table 1: Generation and Load Parameters.

Bus	Generation		Load		V in p.u	Remarks
	P in MW	Q in MVA _r	P in MW	Q in MVA _r		
1	-	-	65	30	1.04	Slack bus
2	0	0	115	60	1.00	Load bus
3	180	-	70	40	1.02	PV bus
4	0	0	70	30	1.00	Load bus
5	0	0	85	40	1.00	Load bus

Table 2: Transmission Line Detail.

Line	Length (km)	Resistance (p.u.)	Reactance (p.u.)	Charging MVAR
1-2	64.4	0.042	0.168	4.1
1-5	48.3	0.031	0.126	3.1
2-3	84.8	0.031	0.126	3.1
3-4	128.7	0.084	0.336	8.2
3-5	80.5	0.053	0.210	5.1
4-5	96.5	0.063	0.252	6.1

This detailed model enables the simulation of various compensation scenarios, allowing for the evaluation of voltage stability, power transfer efficiency, and system resilience under different operating conditions.

Each case study builds upon the previous one, analyzing the network's performance under different conditions and adding incremental improvements

2.1 Case Study 1: Network Simulation (Without Compensation)

Problem Statement:

The primary goal of this case study is to simulate the network's power flow without any compensation. In this initial state, we expect issues related to voltage drops and inefficiencies in power distribution, particularly due to the lack of reactive power support.

Without shunt or series compensation, the transmission lines may experience high reactive power flow, leading to lower voltage stability and higher transmission losses, especially in long-distance lines.

Procedure:

1. Set up the system configuration in PowerWorld Simulator:

- Input the generator values, load demands, and transmission lines as provided in the document.
- Ensure that all buses, generators, and loads are configured correctly.

2. Run the power flow simulation:

- The simulator will show the voltage levels, power generation, and loads across the system.
- Identify any buses with low voltage (below 0.95 p.u.) or excessive power losses.

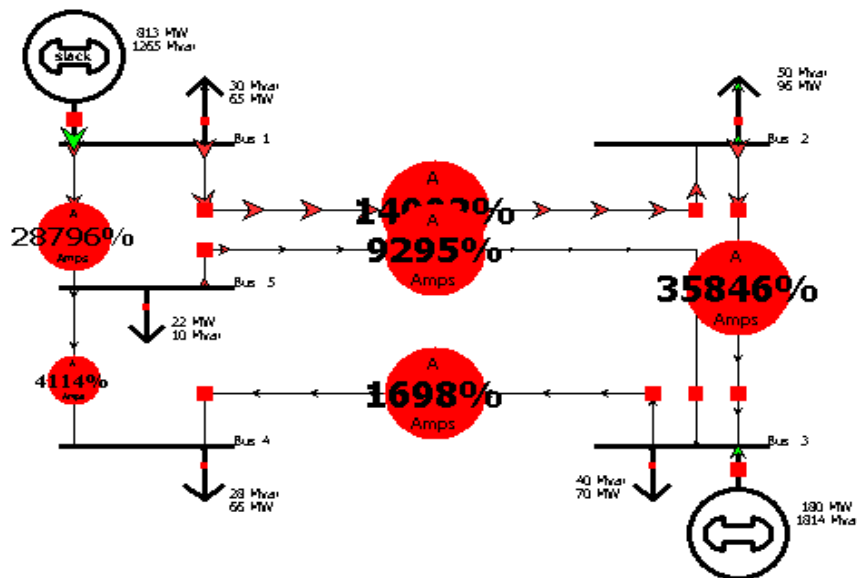
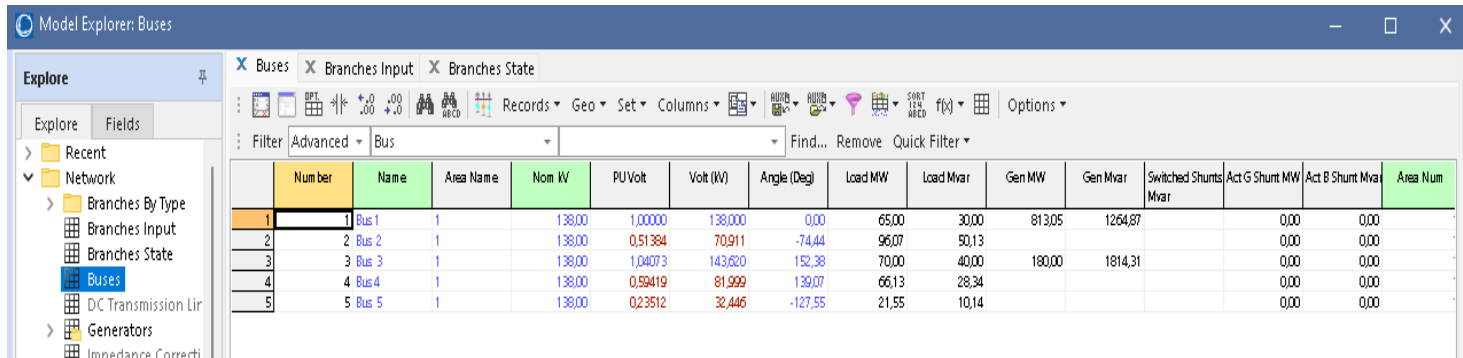


Figure 1: Case 1 Power system

3. Record the results:

- Capture bus voltage, generation (MW & MVAR), and load (MW & MVAR) at each bus.



The screenshot shows the 'Model Explorer: Buses' window. On the left is a tree view with 'Recent' and 'Network' folders. Under 'Network', there are sub-folders for 'Branches By Type', 'Branches Input', 'Branches State', 'Buses' (selected), 'DC Transmission Lin', 'Generators', and 'Impedance Converte'. The main area displays a table with columns: Number, Name, Area Name, Nom kV, PU Volt, Volt (kV), Angle (Deg), Load MW, Load Mvar, Gen MW, Gen Mvar, Switched Shunts Mvar, Act G Shunt MW, Act B Shunt Mva, and Area Num. The table contains 5 rows of data for Bus 1 through Bus 5.

Number	Name	Area Name	Nom kV	PU Volt	Volt (kV)	Angle (Deg)	Load MW	Load Mvar	Gen MW	Gen Mvar	Switched Shunts Mvar	Act G Shunt MW	Act B Shunt Mva	Area Num
1	Bus 1	1	138,00	1,00000	138,000	0,00	65,00	30,00	81,305	1264,87		0,00	0,00	
2	Bus 2	1	138,00	0,51384	70,911	-74,44	96,07	50,13				0,00	0,00	
3	Bus 3	1	138,00	1,04073	143,620	152,38	70,00	40,00	180,00	1814,31		0,00	0,00	
4	Bus 4	1	138,00	0,59419	81,999	139,07	66,13	28,34				0,00	0,00	
5	Bus 5	1	138,00	0,23512	32,446	-127,55	21,55	10,14				0,00	0,00	

Figure 2:Case 1 Model Explorer Results 1

Solution:

The initial simulation without compensation allows for the identification of weak points in the system, particularly buses with low voltage and transmission lines with high losses. This provides a baseline for improvement in future simulations. From the results, we can see which buses would benefit most from shunt or series compensation, where reactive power compensation could improve voltage stability, and where there are bottlenecks in power transfer.

Case conclusion:

This baseline study is crucial for understanding the network's natural operating conditions and identifying specific areas that need intervention. It enables targeted compensation strategies, laying the groundwork for the subsequent case studies.

2.2 Case Study 2: Adding Shunt Compensation

Problem Statement:

Based on the results from Case Study 1, buses with low voltage and high reactive power losses were identified. These issues need to be addressed by adding shunt compensation (capacitor banks) to improve voltage profiles and reduce the reactive power burden on the transmission system. The goal is to inject reactive power directly at the load buses to support voltage.

Procedure:

1. Set up the system configuration in PowerWorld Simulator:

- Input the generator values, load demands, and transmission lines as provided in the document.
- Ensure that all buses, generators, and loads are configured correctly.

2. Identify critical buses:

- Based on the voltage results from Case Study 1, select buses with voltage drops below acceptable levels (0.95 p.u.).

3. add shunt compensation:

- Use capacitors to inject reactive power at these buses.
- Add shunt compensation at appropriate buses using PowerWorld Simulator and specify the size (MVAR) of the capacitor.

4. Run the simulation:

- After implementing shunt capacitors, rerun the power flow simulation and observe the changes in voltage and reactive power flows.

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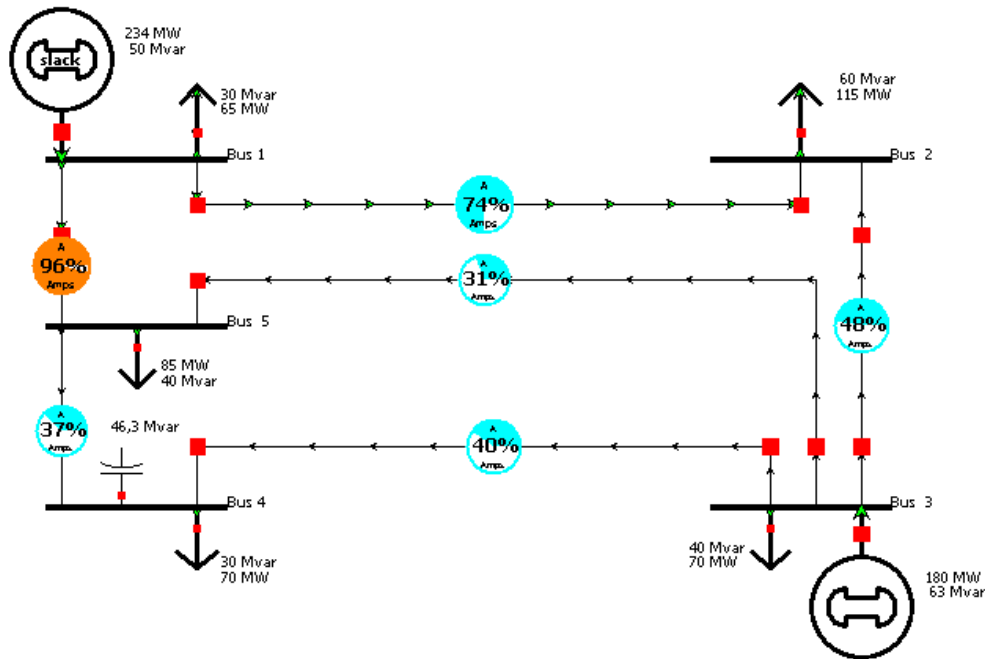


Figure 3: Case 2 Power System

5. Record results:

- Compare the bus voltages and power flows before and after compensation.
Before and after

Model Explorer: Buses

Explore Fields

Recent

Network

Branches By Type

Branches Input

Branches State

Buses

DC Transmission Lin

Generators

Impedance Correcti

Buses

Advanced

Filter

Find...

Remove

Quick Filter

Number	Name	Area Name	Nom KV	PU Volt	Volt (KV)	Angle (Deg)	Load MW	Load Mva	Gen MW	Gen Mvar	Switched Shunts Mvar	Act G Shunt MW	Act B Shunt Mvar	Area Num
1	Bus 1	1	138,00	1,00000	138,000	0,00	65,00	30,00	813,05	1264,87		0,00	0,00	
2	Bus 2	1	138,00	0,51384	70,911	-74,44	96,07	50,13				0,00	0,00	
3	Bus 3	1	138,00	1,04073	143,020	152,38	70,00	40,00	180,00	1814,31		0,00	0,00	
4	Bus 4	1	138,00	0,59419	81,999	139,07	66,13	28,34				0,00	0,00	
5	Bus 5	1	138,00	0,23512	32,446	-127,55	21,55	10,14				0,00	0,00	

Figure 4: Case 1 Model Explorer Results

Model Explorer: Buses

Explore Fields

Recent

Network

Branches By

Branches Inp

Branches Sta

Buses

DC Transmiss

Generators

Impedance C

Line D-FACTS

Line Shunts

Buses

Advanced

Filter

Find...

Remove

Quick Filter

Number	Name	Area Name	Nom KV	PU Volt	Volt (KV)	Angle (Deg)	Load MW	Load Mva	Gen MW	Gen Mvar	Switched Shunts Mvar	Act G Shunt MW	Act B Shunt Mvar	Area Num	Zone Num
1	Bus 1	1	138,00	1,00000	138,000	0,00	65,00	30,00	233,54	49,55		0,00	0,00	1	1
2	Bus 2	1	138,00	0,96721	133,475	-7,21	115,00	60,00			46,77	0,00	0,00	1	1
3	Bus 3	1	138,00	1,00000	138,000	-4,20	70,00	40,00	180,00	63,23		0,00	0,00	1	1
4	Bus 4	1	138,00	0,97610	134,701	-12,28	70,00	30,00			46,30	0,00	0,00	1	1
5	Bus 5	1	138,00	0,95538	131,842	-6,86	85,00	40,00				0,00	0,00	1	1

Figure 5: Case 2 Model Explorer Results

Solution:

The results after adding shunt capacitors should show an improvement in the voltage profile at the selected buses. The reactive power support helps raise the voltage, making the system more stable and reducing losses. Shunt compensation proves effective in mitigating voltage dips, especially in areas far from generation sources. This method of compensation is widely used in power systems to regulate voltage and improve power quality.

Case conclusion:

Shunt compensation is widely used to regulate voltage and enhance power quality in power systems. This case study demonstrates its effectiveness in stabilizing voltage across the network, maintaining power quality, and preparing the system for higher loads.

2.3 Case Study 3: Adding Series Compensation:

Problem Statement:

While shunt compensation improves voltage at buses, series compensation is needed to enhance the efficiency of power transfer through transmission lines. Series capacitors reduce the effective reactance of transmission lines, allowing for higher power transfer capability and reducing line losses.

Procedure:

1. Identify transmission lines with high reactance:

- From the results of Case Study 1 and 2, locate lines with significant power losses or insufficient power transfer capability.

2. Add series compensation:

- Series capacitors are installed on long-distance transmission lines with high impedance.

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- Input the series compensation values in PowerWorld Simulator, specifying the desired level of compensation to reduce line reactance.

Model Explorer: Branches Input

Explore Fields

Recent Network

Branches By:

Branches Input

Branches Sta

Buses

DC Transmis:

Generators

Impedance C

Line D-FACTS

Line Shunts

Branches Input

AC and DC

Buses

Records

Geo

Set

Columns

Find...

Remove

Quick Filter

Options

From Number	From Name	To Num	To Name	Circuit	Status	Branch Device Type	Xfrmr	R	X	B	Lim MVA A	Lim MVA B	Lim MVA C
1	1 Bus 1	2 Bus 2	1	Closed	Line	NO		0,04200	0,16800	0,04100	100,0	0,0	0,0
2	1 Bus 1	5 Bus 5	1	Closed	Line	NO		0,03100	0,12601	0,03100	100,0	0,0	0,0
3	2 Bus 2	3 Bus 3	1	Closed	Line	NO		0,03100	0,12600	0,03100	100,0	0,0	0,0
4	4 Bus 4	3 Bus 3	1	Open	Line	NO		0,08400	0,33600	0,08200	100,0	0,0	0,0
5	4 Bus 4	3 Bus 3	2	Closed	Series Cap	NO		0,08400	0,25200	0,08200	100,0	0,0	0,0
6	5 Bus 5	3 Bus 3	1	Open	Line	NO		0,05300	0,21000	0,05100	100,0	0,0	0,0
7	5 Bus 5	3 Bus 3	2	Closed	Series Cap	NO		0,05300	0,15750	0,05100	100,0	0,0	0,0
8	5 Bus 5	4 Bus 4	1	Closed	Line	NO		0,06300	0,25200	0,06100	100,0	0,0	0,0

Figure 6: Case 3 Model Explorer Branch Parameters

3. Run the simulation:

- Observe how series compensation affects power flow through the transmission lines.

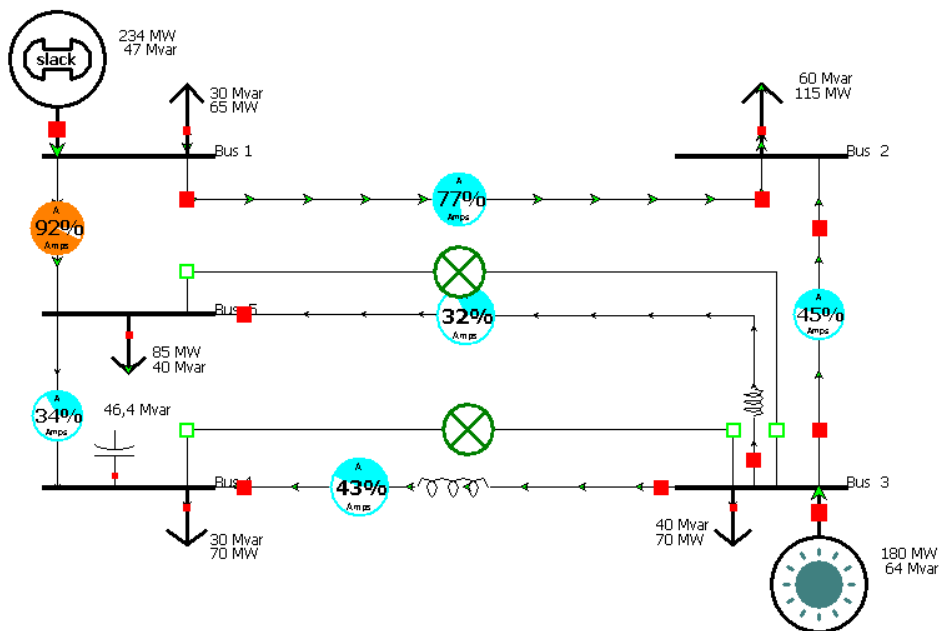
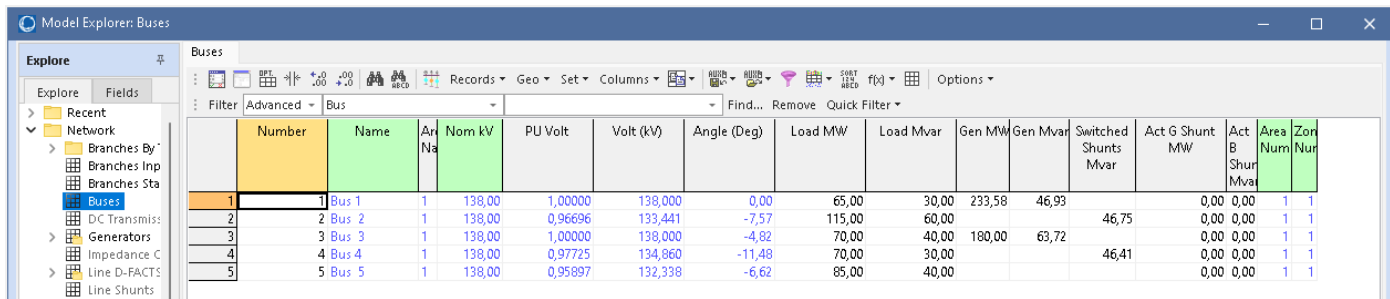


Figure 7: Case 3 Power System

4. Compare results:

- Compare the efficiency of power transfer, voltage stability, and overall system losses before and after series compensation.

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The screenshot shows the 'Model Explorer: Buses' window. On the left is a tree view with 'Recent' and 'Network' folders. Under 'Network', there are 'Branches By' (Inp, Sta), 'Buses', 'DC Transmiss', 'Generators', 'Impedance C', 'Line D-FACTS', and 'Line Shunts'. The 'Buses' folder is selected. The main area displays a table with columns: Number, Name, Ar Na, Nom kV, PU Volt, Volt (kV), Angle (Deg), Load MW, Load Mvar, Gen MW, Gen Mvar, Switched Shunts Mvar, Act G Shunt MW, Act B Shunt Mvar, Area Num, and Zon Nur. The table contains 5 rows of data for buses 1 through 5.

Number	Name	Ar Na	Nom kV	PU Volt	Volt (kV)	Angle (Deg)	Load MW	Load Mvar	Gen MW	Gen Mvar	Switched Shunts Mvar	Act G Shunt MW	Act B Shunt Mvar	Area Num	Zon Nur
1	Bus 1	1	138,00	1,00000	138,000	0,00	65,00	30,00	233,58	46,93		0,00	0,00	1	1
2	Bus 2	1	138,00	0,96696	133,441	-7,57	115,00	60,00			46,75	0,00	0,00	1	1
3	Bus 3	1	138,00	1,00000	138,000	-4,82	70,00	40,00	180,00	63,72		0,00	0,00	1	1
4	Bus 4	1	138,00	0,97725	134,860	-11,48	70,00	30,00			46,41	0,00	0,00	1	1
5	Bus 5	1	138,00	0,95897	132,338	-6,62	85,00	40,00				0,00	0,00	1	1

Figure 8: Case 3 Model Explorer Results

Solution:

Series compensation allows for greater power transfer through long-distance transmission lines by reducing the reactive impedance. This improves overall system efficiency and can prevent overloading of certain lines by allowing more power to flow through others. By improving the transmission line capacity, series compensation reduces the need for additional generation in load areas and helps stabilize the grid.

2.4 Case Study 4: Implementing Renewable Energy Sources

Problem Statement:

With increasing demand for clean energy, renewable sources such as solar or wind can replace or supplement traditional methods of reactive power compensation. Instead of relying on capacitors, renewable energy integration can provide active power generation while indirectly supporting voltage stability.

Procedure:

To choose the best renewable energy solution I had to consider the location which is Tshwane.

The best renewable energy solution for Pretoria, South Africa is solar photovoltaics (PV) systems. Solar PV systems convert sunlight directly into electricity using solar cells. Solar PV systems are clean, reliable, and affordable, making them a good option for both residential and commercial use.

Pretoria has abundant sunshine:

Pretoria receives an average of 2,500 hours of sunshine per year, making it one of the sunniest cities in the world .

This means that solar PV systems in Pretoria can generate a significant amount of electricity.

Even though solar PV is a reliable and excellent renewable energy option for Pretoria, it's important to consider other aspects including the cost of initial installation, the requirement for regular maintenance, and the effect on the regional electrical system. A comprehensive strategy for energy planning could include a variety of renewable energy sources, but with Pretoria's excellent sunlight conditions, solar energy is a popular and sensible option

1. Select a renewable energy source:

- Based on local conditions, I chose a solar generator to integrate into the grid.
- Identify a bus where renewable energy can be added, potentially replacing the need for shunt compensation.

2. Add renewable generation:

- The size of the solar or wind farm into PowerWorld Simulator will be set to 80MW and 40MVAR, specifying the generation capacity (MW) and reactive power output (MVAR).

3. Run the simulation:

- Observe how adding renewable energy affects the overall system, including bus voltages and power flows.

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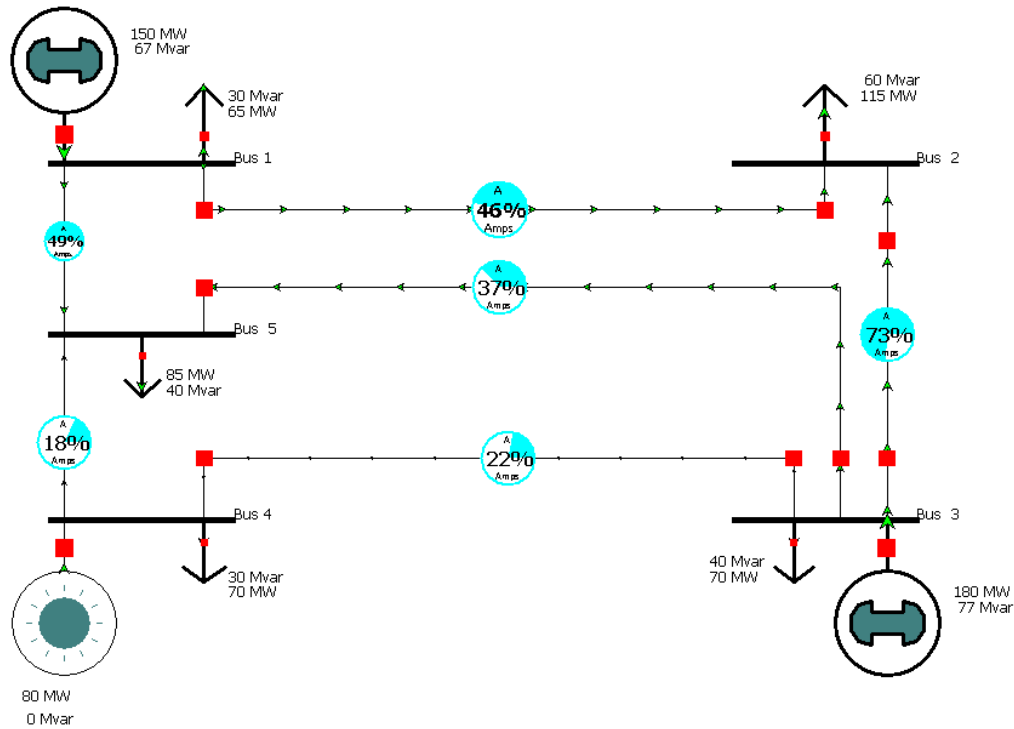


Figure 9: Case 4 Power System

Number	Name	Area	Nom kV	PU Volt	Volt (kV)	Angle (Deg)	Load MW	Load Mvar	Gen MW	Gen Mvar	Switched Shunts Mvar	Act G Shunt Mvar	Act B Shunt Mvar	Area Num	Zone Num
1	Bus 1	1	138,00	1,00000	138,000	0,00	65,00	30,00	149,51	66,90	46,85	0,00	0,00	1	1
2	Bus 2	1	138,00	0,98021	133,585	-4,25	115,00	60,00	180,00	76,52	0,00	0,00	0,00	1	1
3	Bus 3	1	138,00	1,00000	138,000	0,97	70,00	40,00	80,00	0,00	0,00	0,00	0,00	1	1
4	Bus 4	1	138,00	0,93777	129,413	0,52	70,00	30,00	80,00	0,00	0,00	0,00	0,00	1	1
5	Bus 5	1	138,00	0,94938	131,014	-2,42	85,00	40,00				0,00	0,00	1	1

Figure 10: Case 4 Model Explorer Results

4. Cost comparison:

- The costs and benefits of renewable energy integration versus traditional shunt or series compensation, considering both financial and environmental impacts.

Solution:

Renewable energy sources not only generate active power but can also contribute to grid stability. In this case, the implementation of solar or wind power can potentially reduce the need for traditional reactive power compensation, providing a more sustainable and cost-effective solution. The integration of renewables also aligns with global sustainability goals, offering long-term benefits for both the grid and the environment.

3. Importance of Implementing a Comprehensive System for All Case Studies:

The simulations and analyses performed across these case studies highlight the importance of a balanced approach to power system compensation and optimization. By integrating shunt and series compensation alongside renewable energy sources, the system becomes more robust, stable, and efficient. The following benefits are achieved:

- **Improved voltage stability:** Compensation methods help maintain voltage levels within acceptable limits, reducing the risk of voltage collapse.
- **Enhanced power transfer efficiency:** Series compensation allows for more effective power flow across transmission lines, lowering losses.
- **Sustainability and cost-efficiency:** The use of renewable energy not only reduces carbon emissions but also offers a long-term, cost-effective solution compared to traditional reactive power compensation methods.
- **System resilience:** A well-compensated system can handle higher loads and respond to dynamic changes in power demand more effectively, improving overall reliability.

By following these steps and implementing a well-rounded compensation strategy, the system can achieve higher performance, reduced losses, and greater environmental sustainability.

4. Conclusion:

This project has provided a comprehensive analysis of a five-bus power system network, evaluating the performance impacts of various compensation techniques including shunt and series compensation, as well as the integration of renewable energy sources. The findings underscore the critical role that compensation methods play in enhancing voltage stability, improving power transfer efficiency, and supporting the transition towards sustainable energy solutions. Through systematic case studies, the research illustrated how each compensation strategy contributes uniquely to the overall resilience and reliability of the power system. The insights gained from this study not only emphasize the necessity of tailored compensation strategies in

addressing specific system challenges but also highlight the potential for renewable energy sources to play a pivotal role in modern power systems. As the demand for efficient and sustainable electricity delivery continues to grow, the integration of these strategies will be essential for optimizing power system operations and ensuring a stable energy future.

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